

Relaxation Mechanisms and Shales

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ABSTRACT: Unconventional petroleum resources present in shales have recently seen increased development due to growing demand for energy worldwide, declining conventional petroleum discoveries, and improved technology for production. Shale reservoirs differ from more conventional reservoirs in both matrix composition and structure, tending to be low in porosity and ultralow in permeability, making traditional laboratory characterization methods difficult to apply. The noninvasiveness of nuclear magnetic resonance (NMR) makes it a favored technique for shale analysis. However, interpretation of NMR relaxometry results for shales is more complex than in traditional reservoirs. Surface relaxivity in conventional reservoirs is predominantly caused by the interaction of fluid molecules with paramagnetic impurities on pore surfaces. However, in shales, small pore sizes coupled with the presence of hydrogen-rich organic matter can result in more interactions. This article discusses different relaxation mechanisms that may be present in shales and situations where they are relevant. Surface relaxation in organic matter is likely caused by homonuclear dipolar coupling, not paramagnetic impurities, and as such, will have a significant temperature dependence. Due to the small pore sizes in shales, the effect of internal gradients on the signal appears negligible in almost all situations. Analysis of transverse relaxation resulting from organic solids by an inverse Laplace transform may lead to results that are anomalously short and overcall signal intensity. Diffusional coupling between the shale pores confounds interpretation of the NMR response as a pore size distribution. There also appears to be rapid exchange of magnetization between mobile and immobile phases in the samples. © 2014 Wiley Periodicals, Inc. Concepts Magn Reson Part A 43A: 57–78, 2014.

KEY WORDS: porous media; shale; surface relaxivity; internal gradients; dipolar coupling; magnetization transfer

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INTRODUCTION

Accurate understanding of matrix and fluid properties is vital to both the evaluation of and production from a petroleum reservoir. To aid in this characterization, the petroleum industry has been using nuclear magnetic resonance (NMR) to characterize fluid bearing porous materials since shortly after the technology's inception (1–3). A significant advantage of NMR is that it

provides a lithology-independent measure of porosity in the wellbore; all other logging tools produce a result that is formation dependent and incorrect selection of the rock matrix during data analysis can lead to systematic errors in calculated porosity (4,5). As lithology can vary greatly along the length of a well, accurate rock type selection throughout the wellbore can be a significant challenge. In contrast, NMR in conventional reservoirs measures only on the fluids present, such that if the hydrogen indices of the saturating fluids are known, porosity can be directly calculated from the NMR signal (6). Beyond porosity, the relaxation of the NMR signal has been used by the petroleum industry to provide information on pore sizes (7,8), movable and bound fluids (9,10), permeability (11,12), and types of fluids present (13).

The ability to characterize petroleum systems via NMR comes from the enhanced relaxation that occurs when a fluid is placed in a porous material. The NMR signal of bulk water has a relaxation time on the order of a few seconds. Light to medium oils have relaxation times ranging from a few seconds down to tens of milliseconds and are generally a function of viscosity. However, when these fluids are placed in porous materials, such as a petroleum reservoir rock, the fluid molecules will interact with pore surfaces, leading to a decrease in their relaxation time. For longitudinal relaxation, this is described by the relation:

$$\frac{1}{T_1} = \frac{1}{T_{1\text{Bulk}}} + \rho_1 \frac{S}{V} \quad (1)$$

where $T_{1\text{Bulk}}$ is the relaxation time of the bulk fluid, S is the surface area of the pore, and V is the volume of the pore. The surface area to volume ratio is used as a proxy for inverse pore size. Small pores have large surface to volume ratios while larger pores have a small surface to volume ratio. ρ_1 is the surface relaxivity of a pore, which describes how effective the material is at relaxing protons in the fluid. The resulting NMR relaxation rates will be a summation of all the different relaxation times present in the sample. Therefore, when the pore space is saturated with a single phase, the fluid in the small pores will relax more quickly than the fluid in larger pores and the resulting measured distribution of T_1 times can be considered a reflection of the pore size distribution.

The relation describing the transverse relaxation behavior in porous materials is similar to T_1 , but more complicated:

$$\frac{1}{T_2} = \frac{1}{T_2} + \rho_2 \frac{S}{V} + \frac{D\gamma^2 G^2 T_E^2}{12} \quad (2)$$

where γ is the gyromagnetic ratio of the nucleus of interest, D is the diffusion coefficient of the molecules,

G is gradient strength, ρ_2 is the surface relaxivity, and T_E is the echo spacing. Unlike T_1 relaxation, the presence of magnetic gradients in the sample will lead to additional dephasing of transverse magnetization, increasing the T_2 relaxation rate. These gradients may be intended, such as gradients applied to perform diffusion measurements, or may arise naturally from the magnetic susceptibility differences between the pore matrix and saturating fluid. In the case of the naturally arising internal gradients, this effect is generally unwanted, and low frequency magnets and short echo spacings are used to prevent the unwanted influence of internal gradients on the T_2 measurement (14). However, sometimes this effect is used to provide additional information about the system. Decay due to diffusion in the internal field measurements provide information on pore shape and connectivity (15), whereas other work has used internal gradients to provide information about system wettability (16).

In order for the T_1 or T_2 relaxation to be considered a reflection of the sample's pore sizes, the system needs to be in the fast diffusion limit. In this case, the fluid molecules filling a pore will rapidly explore the pore space, but exchange between pores is slow, and therefore, relaxation in a single pore will be mono-exponential. A system is assumed to be in the fast diffusion limit if:

$$\frac{2D}{r} \gg \rho_{1,2} \quad (3)$$

where r is the smallest dimension of the pore space. The other situation, the slow-diffusion or diffusion-limited regime, exists if molecules near the pore surface relax too quickly for diffusion to adequately average the relaxation rate throughout the pore. This occurs when the surface relaxation is extremely high or the pore sizes are very large. A system is assumed to be in the slow-diffusion regime if:

$$\frac{2D}{r} \ll \rho_{1,2} \quad (4)$$

In the case of slow-diffusion, a single pore will have a multiexponential decay. Another issue that arises in porous materials is diffusional coupling. In this situation, there is significant exchange between pores of different sizes, generally between micro- and macropores. Instead of fluid molecules within each pore type relaxing separately, making the measured signal a summation of the different relaxation rates (17), there is considerable exchange of fluid molecules between the pores and the resulting overall relaxation is an average of the pore sizes. Diffusional coupling tends to arise in systems with low surface relaxivities. As such, carbonates often have this issue (18,19).

Further deviation of the measured relaxation behavior from the pore size distribution will occur if the pore space is filled with viscous fluids. For low viscosity fluids, the effect of surface relaxivity can be assumed to dominate over the relaxation that occurs due to intramolecular dipolar coupling within the fluid. However, with more viscous fluids, the effect of internal relaxation begins to compete with the effects of surface relaxation. The dominant relaxation mechanism is a function of the fluid viscosity, pore size, and surface relaxivity. Three regimes can be described. The system can be assumed to be dominated by the internal relaxation of the viscous fluid if:

$$\frac{V}{T_{\text{Bulk}}\rho S} \geq 20 \quad (5)$$

where T_{Bulk} is the T_1 or T_2 bulk relaxation rate for the fluid. For ratios greater than 20, the effect of surface relaxation on the signal is less than 5% and can be assumed to be negligible. Similarly, if

$$\frac{V}{T_{\text{Bulk}}\rho S} \leq 20 \quad (6)$$

the system is dominated by surface relaxation and the effect of internal fluid relaxation is negligible.

In the case where

$$0.05 \leq \frac{V}{T_{\text{Bulk}}\rho S} \leq 20 \quad (7)$$

both internal interactions of the fluid and surface relaxivity will significantly affect the measured relaxation. Note, these relations will not hold for T_1 of very viscous fluids, as the relaxation time may be very long, but there will be no significant interaction between the surface and the fluid molecules due to the extremely low diffusion coefficient.

SHALES

The growing global demand for energy along with diminishing conventional petroleum discoveries has led to increased interest in unconventional petroleum resources. Among the most productive of these unconventional resources are shale resources. By a strict geological definition, shales are clay-rich rocks that show fissile behavior and may contain silt-sized particles of other types of minerals. In the petroleum industry, the term shale is often used more indiscriminately and might be used to describe any system that contains clay to silt-sized particles or requires hydraulic fracturing for hydrocarbon production. As such, reservoirs that primarily consist of silt-sized silicate particles or fine grained carbonates may get labeled as shales.

Shales are different from conventional petroleum systems in several ways. Conventional petroleum reservoirs were considered to be composed primarily of sandstone or carbonates, with high porosity and permeability. The oil and gas present in these reservoirs has migrated in from another location due to buoyancy forces. Usually the presence of clay or organic materials in these reservoirs is minimal. In contrast, many of the shales contain significant quantities of organic matter or clays. This organic matter contains hydrogen and may be visible to many types of NMR equipment, such that the resulting signal from shales can no longer be assumed to be matrix independent. Shales are often self-sourced, meaning the petroleum products in place were produced within the shales through cracking of the organic matter during catagenesis (20). Some of the produced oil and gas may have been expelled, in some cases migrating upward to be captured in a conventional petroleum reservoir, but often considerable amounts are retained in the shale due to the low permeability.

Shales have porosities that are usually significantly lower than conventional reservoirs. Traditional reservoirs have porosity ranging from 10% to 35% whereas shales often have single digit porosities. The pore sizes in shales are typically in the nanometer range, several orders of magnitude lower than in conventional reservoirs, which have pores on the order of microns. Shales may contain a range of different types of porosity, including: interparticle porosity, intraparticle porosity, and organic porosity (21). There may be one dominant type of porosity, such as in the Bossier or Barnett shales, or there may exist a mix of different types of porosity as with the Niobrara shale.

Interparticle porosity exists between the rock grains, clay platelets, or organic matter particles in the shale. Interparticle porosity may contain either water, oil, or gas, though this will be influenced by the wettability of the pore surfaces and capillary pressure of the system. Intraparticle porosity may arise within clay plates, nanopores in minerals, molds, in void spaces from fossils, or within pyrite framboids. Although conventional reservoirs tend to show a clear distinction between the interparticle and intraparticle porosity, it is often not straightforward to distinguish the two types in shales due to the very small pores present. As with interparticle porosity, intraparticle pores may contain oil, water, or gas.

Organic porosity is void space that occurs in organic matter. Because of its composition, organic porosity is frequently hydrophobic, however, some moieties may absorb water through hydrogen bonding (22). Therefore, organic matter porosity is expected to contain primarily oil or gas, but it may contain water under certain

circumstances, such as, after hydraulic fracturing of the reservoir (23). Initially, organic matter starts out with no porosity accessible to water or liquid oil. As the shale undergoes catagenesis, the conversion of organic matter to oil and gas leads to the formation of porosity. The porosity in organic matter will increase with increased thermal maturity, and then above a certain level of maturity, may decrease again. It is believed this loss of porosity is due infill of less viscous forms of organic matter or by collapse of the organic structure. Organic matter porosity may develop as large pendular pores or smaller pores with a spongy texture (24). It should be noted, pyrite framboids often appear in conjunction with organic matter. This is because the reducing environment necessary for development of pyrite is also conducive to the preservation of organic matter, which can often be found within the intracrystalline porosity of the pyrite framboids or adjacent to them.

The small pore sizes in shales often leads to extremely low permeability. This hinders traditional methods of core analysis (21), which often require liquids or gases to flow through the samples. Because NMR is noninvasive, it is a popular method to try to characterize these ultratight porous materials. Some work has been done using high resolution spectroscopy to characterize shales (25–28), but relaxometry is the primary NMR method used. However, interpreting the resulting relaxation time distributions for shales is more difficult. Shale NMR measurements are often analyzed using the standard relaxation behavior described in Eqs. (1) and (2). However, in systems with such small pores, there are several types of interactions that may arise in shale samples that are negligible in the large pore sizes of conventional sandstone and carbonate reservoirs. Currently, the effect of these mechanisms are all classed as surface relaxivity but grouping these interactions into the single value may lead to loss of valuable information at best or incorrect interpretation of the NMR results at worst. This article revisits possible relaxation mechanisms that could arise in shales and discusses their impact on the measured signals.

TRADITIONAL SURFACE RELAXIVITY

A great deal of work has been devoted to understanding the surface relaxivity mechanism in sandstones and carbonates. Korrington, Seevers, and Torrey (KST) (29) put forward the idea that relaxation at pore surfaces occurs through interactions of fluid molecules with paramagnetic impurities. Their model assumes that diffusion of fluid molecules is rapid enough that the distribution of the impurities is unimportant, only their quantity. They

describe two time scales: τ_s , which describes the migration within the fluid layer at the surface and τ_c , which describes the exchange rate between molecules on the surface and bulk fluids. For the assumption that the exchange of fluid molecules between the surface and the bulk fluid is more rapid than their diffusion rate across a pore surface, or $\tau_s \gg \tau_c$, the description of longitudinal relaxation is:

$$\frac{1}{T_1} = \frac{1}{T_1^0} + \frac{\frac{N_s}{N}}{T_s^0 + \tau_c} + \frac{\frac{N_p}{N}}{T_p^0 + \tau_c} \quad (8)$$

where T_1^0 is the relaxation rate of the bulk fluid, T_s^0 is the relaxation rate of fluid molecules at the surface not containing paramagnetic impurities, T_p^0 is the relaxation rate of fluid molecules at the paramagnetic impurities, N_p is the number of paramagnetics at the surface, N_s is the number of surface sites not containing paramagnetics, and N is the number of spins in the bulk fluid. While Korrington et al. assumed this situation to be the case, we also present their second case for completeness, where diffusion at the surface is rapid compared to exchange with the bulk, $\tau_s \ll \tau_c$. Here the relaxation is described by:

$$\frac{1}{T_1} = \frac{1}{T_1^0} + \frac{\frac{N_s + N_p}{N}}{T_{s+p} + \tau_c} \quad (9)$$

where

$$\frac{1}{T_{s+p}} = \frac{1}{T_s^0} + \frac{\frac{N_p}{N_s}}{T_p^0 + \tau_s} \quad (10)$$

The KST model accounts for both relaxation arising from paramagnetic and diamagnetic sources. This model was adapted and simplified by Kleinberg et al. (30). Previous work showed the nonmagnetic relaxation effects on fluid in sandstones to be quite weak (31,32), such that they simplified the equation to only account for the effect of paramagnetic impurities:

$$\frac{1}{T_1} = \left(\frac{Sh}{V} \right) \left(\frac{n_m}{T_p + \tau_c} \right) \quad (11)$$

where h is the thickness of the surface layer and n_m is the proportion of the surface occupied by paramagnetic ions.

It is expected that τ_c depends significantly on the temperature, as the rate of exchange of fluid molecules between surface and bulk would increase with temperature. However, when Latour et al. (33) performed numerous measurements of relaxation time as a function of temperature for water saturated rock, they found that for over a range of 25–175°C, the relaxation response was virtually independent of temperature. In

order for this to hold, T_p must dominate such that the influence of the rate of exchange between the surface and bulk is negligible. Therefore, it was surmised that the amount of surface paramagnetic impurities was the controlling factor in surface relaxation in sandstones.

KST did not mention T_2 in their work, however Kleinberg and Horsfield (14) found that for short echo spacings and low magnetic field strengths, relaxation due to diffusion through magnetic inhomogeneities could be neglected and KST could be adapted for T_2 .

$$\frac{1}{T_1} = \left(\frac{Sh}{V} \right) \left(\frac{n_m}{T_p} \right) \quad (12)$$

The effects of specific paramagnetic types on the NMR relaxation rate have been investigated by Keating and Knight (34–36).

Surface relaxivity is more complex in carbonates. Fordham and coworkers (18) found that while temperature independence held for sandstones, a slight increase in relaxation time was observed for some carbonates. Godefroy et al. (37) found minor fluctuations in relaxation times with temperature as well. They speculated that τ_s could not be assumed to be very much larger than τ_c . However, their findings still supported the assumption that relaxation arose from interactions of fluid molecules with paramagnetic centers. Straley (38) also found an increase in T_2 with increased temperature in carbonates, but he proposed another mechanism for the temperature dependence. He saturated carbonates with brine mixtures of either pure H_2O or a small amount of H_2O mixed into D_2O and observed the effect on the relaxation times. Provided surface relaxation stemmed from paramagnetic impurities, the relaxation times should have remained unchanged. However, because the strength of the dipolar interaction between hydrogen and deuterium is significantly lower than between two hydrogens, if the dipolar interaction is a significant source of surface relaxation, the measured relaxation times should increase, which is what Straley observed for some samples.

For shales, a problem arises with a model of surface relaxivity that relies on paramagnetic impurities when trying to reconcile it with their compositions. Because the relaxation time gives a relative pore size distribution, researchers commonly try to calibrate the distributions by comparing the NMR results with pores sizes determined using other methods, such as focused ion beam scanning electron microscopy (FIB-SEM) or mercury injection capillary pressure, to obtain a surface relaxivity value. Several researchers have attempted to do this in shales and have found surface relaxivity values for organic porosity on the order of 40–50 $\mu m/s$ (39,40). These values for surface relaxivity are strikingly high. For comparison, Matteson et al. (41) found

surface relaxivity for glauconite, an iron rich clay, to be 3.3 $\mu m/s$, a value which was later corroborated by work done by Hossain et al. (42). Note, it is possible that the extremely high shale surface relaxivity values are simply an artifact; Hossain et al. (42) compared NMR surface relaxivities calculated from different methods and found that surface relaxivity determined from image analysis methods tended to overestimate surface relaxivity due to underestimation of the surface area caused by insufficient resolution. However, other work where surface relaxivities have been calculated from SEM images show more conventional values (43).

To calculate the weight percent of paramagnetics necessary to produce a given surface relaxivity, we turn to Foley et al. (44). They found that surface relaxivity increased proportionally with the surface concentration of paramagnetic species. The fractional concentration of the paramagnetics by mass is described as:

$$n_M = M_p \frac{MW_{Matrix}}{MW_p} \quad (13)$$

where n_M is the fractional composition by weight, M_p is the concentration of the paramagnetic by weight, MW_{Matrix} is the molecular weight of the matrix, and MW_p is the molecular weight of the paramagnetic. Kerogen has no fixed molecular weight, but has a range of weights upward of 1,000 daltons, which was used for calculation here. Based on their results, to produce a surface relaxivity of 50 $\mu m/s$, the surface concentration of paramagnetics must be approximately 3% and 5% Fe and Mn, respectively, for a quartz-based system, 4% and 8% Fe and Mn for a carbonate-based system, 6%–14 % Fe and Mn in a calcium silicate system, and 36%–60% Fe and Mn in a kerogen system. The concentration number for kerogen appears suspiciously high, indicating that a weight percent calculation is not appropriate here as we are dealing with both micro- and macromolecules. Instead a volumetric approach is more suitable. Here

$$n_v = \frac{M_p \times \rho_p^{-1}}{M_k \times \rho_k^{-1} + M_p \times \rho_p^{-1}} \quad (14)$$

where n_v is the fractional composition by volume, M_p and ρ_p are the concentration by weight and density of the paramagnetic ion and M_k and ρ_k are the concentration by weight and density of kerogen. When we instead calculate the volume percentage of paramagnetics in kerogen, we obtain more realistic appearing values of 1.7% for Fe and 3.5% Mn.

Based on these values, several issues arise. First, trace mineral analysis indicates that Mn exists at concentrations ranging from parts per million to fractions of a percent for organic rich shales and is virtually

nonexistent in the organic matter itself (45). While iron may exist at significant percentages within shales, much of this is associated with pyrite; very little iron is located within the organic matter (45,46). Pyrite tends to form cubic crystals, often growing to a significant size of microns or even millimeters. Therefore, for a shale containing 5% iron, the majority of that is likely to be concentrated in the pyrite crystals with only a small fraction of the iron present distributed as paramagnetic impurities on pore surfaces. While in shales, pyrite framboids may contain significant amounts of intracrystalline porosity, on the order of 20%–30% of the framboid volume, fluid molecules would need to come directly in contact with the pyrite surface to produce any significant effect on relaxation (30). How often this sort of situation arises is uncertain, but FIB-SEM work (47) indicates, it is probably unusual; most framboids appear to be filled with organic material such that fluids would be unlikely to ever directly contact the pyrite surface. As such, alternative explanations are needed to describe the source of the surface relaxivity in shales, particularly in organic matter. Even if the surface relaxivities of 50 $\mu\text{m/s}$ are in fact erroneously high, paramagnetic impurities still appear unlikely to be a significant source of surface relaxivity due to their extremely low concentrations in kerogen and bitumen. This also suggests that significantly different relaxation rates may arise in fluids located in organic porosity versus those in inorganic porosity, potentially hindering the interpretation of the NMR signal as a reflection of the pore size distributions in shales containing mixed type porosity.

INTERNAL GRADIENTS

The presence of internal gradients is known to confound transverse relaxation measurements due to additional, unwanted dephasing of the spin coherence. Although the effects of internal gradients in shales have not been directly addressed previously in the literature, it is possible to infer their influence using results from studies on conventional resources. De Swiet and Sen (48) and Hurlimann (49) have put forth treatises on the important length scales in the pore space. The first of these is the diffusion length, $l_D = \sqrt{D_0 \tau}$, the second is l_s , which is the size of the pore and the third is the dephasing length $l_g = \left(\frac{D_0}{\gamma G}\right)^{1/3}$. The dominant length scale determines how internal gradients will effect a spin echo decay measurement.

The free diffusion regime will always dominate at very short times. The spins have not diffused far enough to experience a significant dephasing nor

encounter the pore surface. In this regime, the effect of gradients on the spin echo decay is described by:

$$\frac{M(\tau)}{M_0} = \exp\left(-\frac{2}{3}\left(\frac{l_D}{l_g}\right)^6\right) \quad (15)$$

where $\frac{M(\tau)}{M_0}$ is the attenuation effect on the signal. In the case where $l_D \ll l_g$, the ratio of $\frac{M(\tau)}{M_0}$ will tend toward 1, and there will be no additional attenuation of the echo decay due to internal gradients.

The next regime is the motional averaging regime. This is the regime where the smallest length scale is the pore size. The modification of the spin echo decay is described by:

$$\frac{M(\tau)}{M_0} = \exp\left(-\frac{1}{60}\left(\frac{l_D}{l_g}\right)^2\left(\frac{l_s}{l_g}\right)^4\right) \quad (16)$$

Again, if l_s is the smallest length scale, the relation of $\frac{M(\tau)}{M_0}$ will also tend toward 1, and there will be no attenuation due to internal gradients.

The last regime is the localization regime. Under this condition, the fluid molecules diffuse long enough to experience a dephasing of 2π radians.

$$\frac{M(\tau)}{M_0} = 5.88\left(\frac{l_g}{l_s}\right)\exp\left(-1.02\left(\frac{l_D}{l_g}\right)^2\right) \quad (17)$$

The localization regime is the only regime where the internal gradients will have a significant impact on the resulting NMR signal.

Assuming diffusion coefficients at room temperature of $2.3 \times 10^{-5} \text{ cm}^2/\text{s}$ for water, $0.5 \times 10^{-5} \text{ cm}^2/\text{s}$ for light oil, and $6.0 \times 10^{-4} \text{ cm}^2/\text{s}$ for gas, we calculate l_g for gradient strengths ranging from 1 to 10^8 G/cm , shown Fig. 1(a). While the resulting range of l_g values are on the order or smaller than the pore sizes in many conventional rocks, even at extreme internal gradient values, l_g is larger than the average pore size of many shales.

We also estimate the potential internal gradients that may arise in the shale samples. While the internal gradients present are not only dependent on pore size but also the pore shape, the magnitude of internal gradients can roughly be described by:

$$G \propto \frac{\Delta\chi B_0}{r} \quad (18)$$

where $\Delta\chi$ is the magnetic susceptibility difference between the pore surface and saturating fluids, B_0 is the strength of the applied magnetic field, and r is the pore size. We used the estimated magnetic susceptibilities of frequent shale constituents from common literature values (50). B_0 was set to 500 Gauss, the

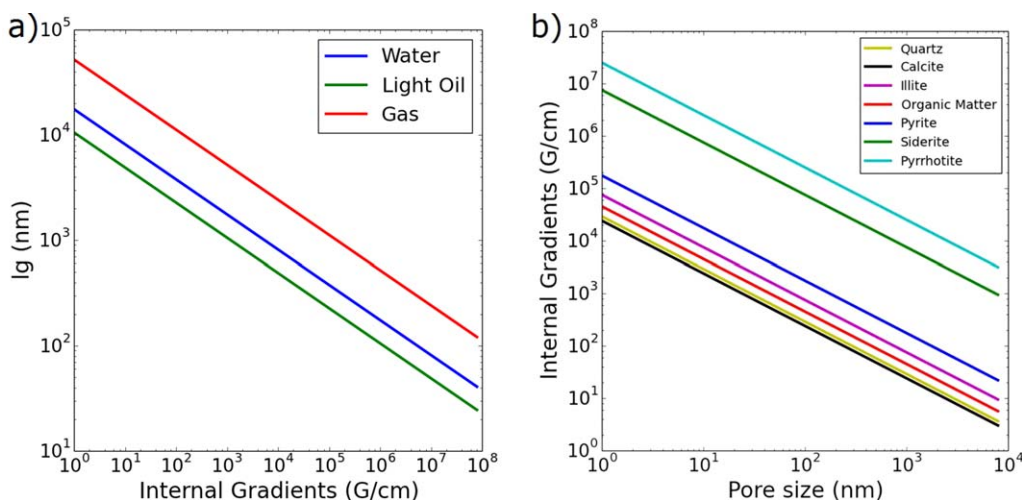


Figure 1 (a) l_g values for a range of gradient strengths for water, light oil, and gas at room temperature. (b) Calculated internal gradients for a range of common shale material constituents.

common field strength for logging and lab instruments used on porous materials. In addition, the magnetic susceptibility may be significantly different depending on the depositional environment of the shale. Marine shales tend to codeposit with paramagnetic impurities such as pyrite, whereas this is much less common for lacustrine shales (51). Based on the results shown in Fig. 1(b), we can see that shale reservoirs are likely to be in the free diffusion or motional averaging regime; given the average pore size and possible gradients in shale reservoirs, it is unlikely that the dephasing length will be dominant. The exception is if there are pores located in siderite or pyrrhotite.

The implication of these results are twofold. While the effect of internal gradients on the T_2 measurement is expected to be minimal, this also indicates that the gradients for diffusion measurements during logging are also unlikely to be effective; the same l_g parameter that is relevant for internal gradients is also applicable for the stray field gradients used for diffusion measurements in logging tools. If we assume the fast diffusion limit, logging tools with gradients on the order of 20 G/cm are unable to see diffusion in pores under $\sim 2 \mu\text{m}$. Pulsed field gradients also run into challenges in measuring diffusion in shales. Unlike the stray field diffusion measurements where a constant gradient is present during the entire measurement, pulsed field gradient measurements need time to ramp the gradients up and down, as well as allow for a period of time for the molecules to diffuse. Depending on the NMR equipment, the time required for this is usually a few milliseconds at minimum. However, in many of the shales, this is on the order of the relaxation time of the system. In both cases, if the molecules move far enough to experience

appreciable attenuation, it is likely they have moved through multiple pores and the fast diffusion limit conditions will not be met, raising questions as to the meaning of the T_2 result, which is discussed further in the following sections. Potentially, some of the gradients available with commercial high-field systems or home built gradients may be strong enough to allow reliable diffusion measurements in shales.

DIFFUSIONAL COUPLING

Another concern in shale samples is diffusional coupling between pores, a known issue in carbonates. There, the pore systems often consist of larger macropores connected to smaller microporosity. The surface relaxation is often quite low in carbonates, such that the diffusion of fluid molecules between the micro- and macropores is rapid enough to average the relaxation times between the pores (18). Anand and Hirasaki (52) put forth a diffusional coupling parameter, α to describe the state of coupling between micro- and macropores.

$$\alpha = \frac{\rho\beta L_2^2}{DL_1} \quad (19)$$

where ρ is the surface relaxivity, β is the fraction of microporosity in the system, L_1 is the half width of the micropore, L_2 is the half length of the macropore, and D is the diffusion coefficient. When α is less than 1, the two systems are in complete coupling such that a single relaxation time is produced from the two populations. For the range $1 < \alpha < 250$, the pores are in an intermediate coupling regime, where two distinct peaks will still arise, but they will not be representative of the

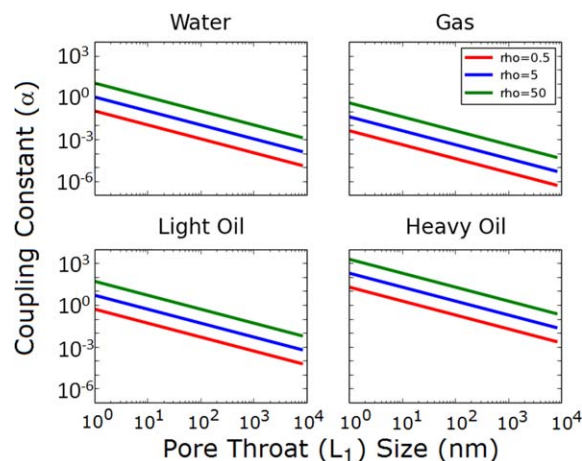


Figure 2 Coupling constants for water, gas, light oil, and heavy oil for a range of low, medium, and high surface relaxivities as a function of pore throat size, L_1 . α values below 1 are considered to be completely coupled. α values above 250 are considered decoupled. α values between 1 and 250 are considered to be in an intermediate coupling region.

porosity fractions. For α greater than 250, the two pores will be decoupled.

We have adapted this convention for nanoporous materials. While work is ongoing to characterize nanoporosity in organic matter, work with FIB-SEM indicates there may be significant connectivity in the nanoporosity in organic matter for some stages of shale maturity (53).

In the case of shales, the diffusional coupling will likely be between pores of different size, but closer in magnitude than for pores in carbonates. In our model, we assume that L_1 , which was the half-width of the micropore in carbonates, is instead equivalent to the throat between the nanopores in shales, an equal fraction of the two nanopore populations and a half-length of the larger nanopore, L_2 , to be 1,000 nm. This is chosen as a conservative upper limit on the size of pores found within shales; most pores are significantly smaller (53). A smaller half-length will lead to increased coupling between the two pores, and hence smaller value of α , while a larger half-length leads to less coupling between the pores. This is because as pore size increases, a smaller fraction of the fluid in the larger pore will exchange with the smaller pore, decreasing the effect of averaging on the relaxation. The results were simulated for low, medium and high surface relaxivity systems using the diffusion coefficients of water, gas, light oil, and heavy oil at ambient conditions, shown Fig. 2. Results are for pores saturated with a single phase. Coupling effects for two or three phase saturations will likely require molecular

dynamics simulations to adequately describe fluid behavior in the pore space.

The results suggest that diffusional coupling will lead to significant averaging of the NMR signal in most situations. For gas, all situations display significant coupling between the pores. With the exception the high surface relaxivity case, water had the values had α below 1 even for the smallest pore throats. In the case of water in a high surface relaxivity material, the α calculated was on the low end of the intermediate coupling range. Light oil was in the intermediate region for pore throats below 10 nm, but mostly in the coupled regime. Only the heavy oil in a high surface relaxivity material with pore throats below 10 nm reached the decoupled regime. This suggests that for many shale systems, the T_2 relaxation time is not a reflection of the pore size distribution, but an average of the pore sizes that is influenced strongly by pore shape and connectivity.

HOMONUCLEAR DIPOLAR COUPLING

Unlike conventional reservoirs, many shale reservoirs are rich in organic materials such as kerogen and bitumen (20,54). Kerogen is a geopolymer created from organic debris preserved during deposition and has been transformed during diagenesis from the heat and pressure of burial. Kerogen structure is different depending on whether its depositional environment was lacustrine, marine, or terrestrial. In addition, as kerogen produces oil and gas during catagenesis, its structure will be altered as aliphatic groups break off, becoming depleted in hydrogen as its thermal maturity increases. It is functionally separated from another type of organic matter, bitumen. Kerogen and bitumen are distinguished as the latter is soluble in most organic solvents while the former is not. Trace amount of bitumen may be present in immature source rocks. However, the majority of bitumen in shales arises as an intermediate product as kerogen cracks into hydrocarbon during catagenesis.

Several researchers have indicated that relaxation in shales may arise from homonuclear dipolar coupling between the fluids and organic material (55,56) While both homonuclear dipolar coupling and interactions with paramagnetics serve to enhance T_1 and T_2 relaxation rates, there are important differences in the behavior of the surface relaxivities. The homonuclear dipolar interaction is described by

$$D_{ij} = -\frac{\mu_0 \gamma^2 \hbar}{4\pi r_{ij}^3} \left(\frac{3\cos^2\theta_{ij} - 1}{2} \right) \quad (20)$$

where D_{ij} is the dipolar coupling constant, μ_0 is the permeability of a vacuum, $\hbar$ is Planck's constant, r_{ij} is the

distance between the two interacting spins, and θ_{ij} is the angle between the applied magnetic field and the line connecting the two spins. As such, the strength of a dipolar interaction depends on the angle the two coupled spins make with the applied magnetic field. Given the maximum value of $\cos^2\theta$ is 1, the maximum possible dipolar coupling constant is then

$$D_{\text{Max}} = -\frac{\mu_0 \gamma^2 \hbar}{4\pi r_{ij}^3} \quad (21)$$

The effect of dipolar coupling on the T_1 relaxation is given as:

$$\frac{1}{T_1} = -\frac{3}{10} \frac{\gamma^4 \hbar^2}{r_{ij}^6} \left(\frac{\tau_c}{1 + \omega_0^2 \tau_c^2} + \frac{2\tau_c}{1 + 4\omega_0^2 \tau_c^2} \right) \quad (22)$$

The effect of the dipolar interaction on the T_2 relaxation time depends on the motional regime. Bloembergen et al. (57) defined three regimes. For the case $\omega_0 \tau_c < 1$, where ω_0 is the frequency of the applied magnetic field and τ_c is the correlation time, we are in the motional narrowing regime, where T_2 is dominated by T_1 and there will only be minimal additional impact on the relaxation rate compared to T_1 . For $\omega_0 \tau_c \gg 1$, we are in the rigid lattice approximation and the T_2 relaxation rate is proportional to the dipolar coupling:

$$\frac{1}{T_2} = -\frac{\mu_0 \gamma^2 \hbar}{4\pi r_{ij}^3} \quad (23)$$

The intermediate case where $\omega_0 \tau_c > 1$, the relaxation is described by the more complex behavior

$$\frac{1}{T_2} = \sqrt{2\pi} \left(\frac{9\pi \gamma^4 \hbar^2}{5 r_{ij}^6} \tau_c \right) + \frac{1}{2T_1} \quad (24)$$

as it also contains a contribution from the T_1 component. Unlike in paramagnetic surface relaxivity, where the rapid electron relaxation rate dominates over τ_c , the relaxation rate of dipolar induced surface relaxation will be highly dependent on the correlation time and thus, system temperature. It is then expected that, due to decreased residence time of fluid molecules, relaxation times will increase with increased temperature, which has been observed (43,56). There exists two possible types of homonuclear dipolar interactions in the shale systems. One is the intermolecular, where a hydrogen bearing fluid molecule encounters a hydrogen atom on a pore surface. The other is intramolecular, where two hydrogens within the same molecule will experience dipolar coupling if molecular rotation is hindered, for example, by adsorption to the pore surface. Depending on the distance between the two hydrogen atoms, the dipolar coupling may range from 10 kHz up to around 100 kHz. Assuming the extreme

case of the rigid-lattice approximation, the maximum relaxation time due to dipolar coupling alone will be on the order of tens of microseconds.

The precise residence time of gas and liquid molecules at pore surfaces in shales is still poorly understood and will likely vary significantly depending on the composition and temperature of the shale. Work with benzene in zeolites at room temperature produced correlations times on the order of $10^{-3} < \tau_c < 10^2$ s (58), indicating that situations where the fluid molecules in shales are in the rigid-lattice approximation may be common. Another potential tool to better understand the dynamics of gas within in shale is hyperpolarized ^{129}Xe , which has been used in a variety of nanoporous materials such as zeolites (59,60), organic rich soils (61), clays (62), and coal (63). Adsorbed xenon experiences a chemical shift (64), allowing it to be distinguished from xenon in the gas phase and potentially providing information as to its residence time on the pore surface. Xenon is both of similar molecular size and wettability (65) to methane, making it a useful analogue. Depending on the surface and temperature, xenon may have surface residence times ranging from nanoseconds to seconds, or possibly even longer, which indicates a similar wide range of residence times may be seen for fluid molecules in shales. Xenon has also been used in exchange experiments between pore space and a dissolved phase (66), which could be used to give information on the behavior of methane dissolved in bitumen and kerogen.

Another impact of relaxation caused by homonuclear dipolar coupling is that the magnetization of the spin can no longer be considered “killed” when it approaches a relaxation site on the surface, as it could with paramagnetic induced relaxation. In addition, the sign of the dipolar coupling of the surfaces in a pore can be either negative or positive depending on the orientation of the pore compared to the applied magnetic field. A spin-bearing fluid molecule that encounters a positive followed by a negative dipolar coupling may “undo” some of the initial dephasing. This means that KST may not be the best model for describing the behavior of fluids within shales. Instead, a model that allows multiple encounters with relaxing sites is likely to be a better description of the system. In this situation, even without diffusional coupling, the T_2 may no longer relate to the size of the pores. If all the fluid molecules have an equal probability to explore the pore space before relaxing, the relaxation of fluid within the pore should still reflect the surface to volume ratio. However, if some molecules are more likely than others to remain near to the surface, then some molecules will preferentially relax over others. A good example where this may arise is in wet gases. Ethane

has a significantly higher adsorption energy than methane, such that it is reasonable to expect it would preferentially be in contact with the pore surface.

Beyond an increased importance of temperature on the NMR response, a system where the relaxation is dominated by homonuclear dipolar coupling means there is the potential for reservoirs where there exists anisotropy in the NMR signal. There is evidence in other porous systems that molecules preferentially align along surfaces when adsorbed (67,68). Adsorption of fluids on clays also shows ordering (69) with nonrandom tumbling of molecules close to the clay surface (70). If a system were to have no preferential orientation of the fluid molecules at pore surfaces, then the dipolar coupling will simply be an average over all angles and no anisotropy in the relaxation will occur.

The dependence of relaxation on orientation with the applied magnetic field has been seen in biological systems (71,72) and elastomers (73). In order for relaxation anisotropy to arise in shales, they would need to be highly ordered, which has been observed (21). Clay particles tend to align parallel to the bedding plane, perpendicular to the direction of maximal stress. This also raises a concern of different NMR responses between logging and lab tools, as a variety of NMR configurations exist. Most logging tools have the applied magnetic field into the formation. In contrast, the MRIL logging tool is a two-dimensional dipole, such that the field is neither purely radial nor azimuthal in the sensitive region. In the lab, superconducting magnets have their applied magnetic field parallel to the bore of the magnet while yoke-type and Halbach magnets have their applied magnetic field perpendicular to the bore. As a pedagogical illustration of the effect of different magnetic field orientations on the relaxation rate, we present a simplified example of a two-dimensional pore.

We assume a rhombus-shaped pore and calculate the possible values of the relaxation angular dependence term $\frac{3\cos^2\theta-1}{2}$ for a range of pore axis aspect ratios (1:1, 1:1.25, 1:2, 1:3, 1:10) depending on their angle with the applied magnetic field. These are calculated for both the fast diffusion regime and the slow diffusion regime situations. In the fast diffusion limit, the relaxation is slow compared to the diffusion rate and fluid molecules will experience all the pore surfaces before relaxing, and therefore positive and negative dipolar coupling constants will average out to a single value for the pore. However, in the slow diffusion limit, where the rate of averaging over the pore is slow compared to the relaxation rate, a fluid molecule at the surface is only likely to experience the dipolar coupling of a single pore surface. Because a dipolar coupling of +10 kHz will have the same effect on the relaxation

rate as a coupling of -10 kHz, and we are concerned with the rate of relaxation, to account for the lack of averaging over the pore space on the relaxation in the slow diffusion limit, we take the absolute value of the angular dependence term.

For pores in the fast diffusion limit, Fig. 3(a), the resulting dipolar coupling constant of a square pore is independent of the angle it makes with B_0 . Even a slight departure from isotropy (aspect ratio 1:1.25) gives a relaxation angular dependence term that is 50% stronger with the long axis aligned parallel to the applied magnetic field than perpendicular to it. The effect rapidly increases as the pore becomes more oblong. As pore anisotropy continues to increase, the dipolar coupling constant becomes negative as the major axis of the pore becomes perpendicular to the applied magnetic field. While the absolute difference between the two relaxation angular dependence terms is large, practically their difference will only be a factor of two. Therefore, the largest effect on the relaxation time with angle will be between when the major axis of the pore is aligned along the applied magnetic field and when it is oriented such that the angular dependence term is crossing from positive to negative.

The case of the slow diffusion limit is more complicated, shown Fig. 3(b). The square pore no longer has a relaxation angular dependence term that is independent of its angle with B_0 . When an axis is aligned along the applied magnetic field, its relaxation angular dependence term is near 0.25. As the axis of the pore approaches an angle of 45° with B_0 , it reaches a value of 0.75, before decreasing again to 0.25. As the pores become more oblong, the relaxation angular dependence term becomes relatively similar over a range of angles. For extremely anisotropic pores, the relaxation angular dependence term will decrease to a minimum, before increasing again to larger values when the major axis of the pores is aligned perpendicular to the applied magnetic field.

Even for a system where the pores are highly oblong, if the pores are completely randomly oriented, the resulting measured dipolar coupling constant will be $\frac{1}{4}D_{\text{Max}}$ for the fast diffusion case and $\frac{1}{2}D_{\text{Max}}$ for the slow diffusion case, regardless of the applied magnetic field orientation. If there is some order to the system, then orientational anisotropy will begin to arise. However, even in a highly ordered system, there will be some variation in the pore orientation. We assume systems of weakly, medium-weak, medium-strong, and strongly ordered pores, shown in Fig. 3(c), and calculate the average relaxation angular dependence term for the different aspect pores for an applied magnetic field both parallel and perpendicular to the bedding plane, shown in Fig. 4 for the fast diffusion limit and Fig. 5 for the slow diffusion limit.

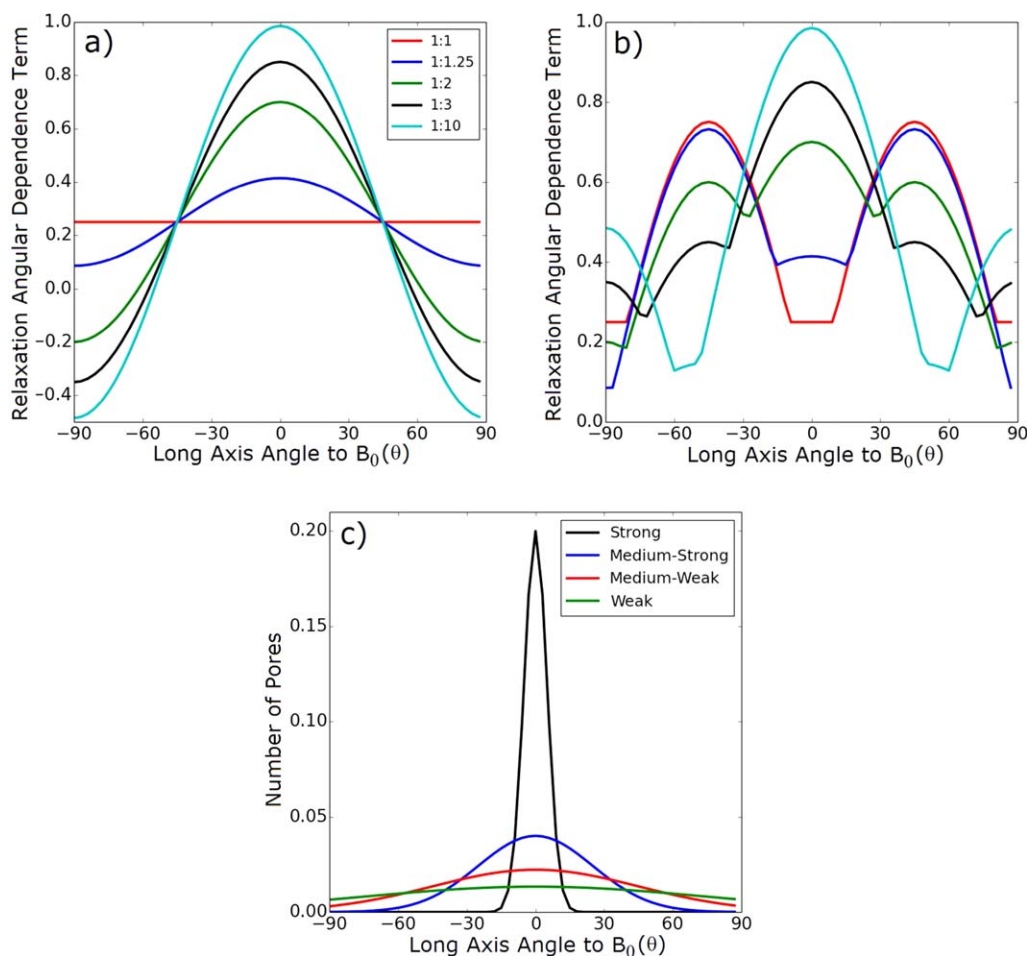


Figure 3 Change in the relaxation angular dependence term as a function of angle to applied magnetic field for a range of aspect ratios in (a) the fast diffusion limit and (b) the slow diffusion limit. The long axis is aligned parallel along the applied magnetic field at 0° and perpendicular at $\pm 90^\circ$ (c) Distribution of angles of the major pore axis for a strongly, medium-strong, medium-weak, and weakly ordered system.

For the case of fast diffusion, we see that while the highly ordered situation leads to the largest absolute difference in the relaxation angular dependence term, because the effect of positive and negative dipolar coupling will produce the same dephasing, the largest effect on the relaxation appears to be observed in the medium-ordered situations. These situations of mediumly ordered porosity are highly probable in realistic shales, raising a significant concern regarding the effect of anisotropy on measured transverse relaxation times. For the case of slow diffusion, the effect of orientation on the relaxation angular dependence term is less dramatic. For highly anisotropic pores, at most a factor of two is seen in the term depending on the orientation of the magnetic field. A more significant effect is seen for some of the less highly anisotropic pores, which display some of the lowest relaxation angular dependence terms of the systems. However, at most

there seems to be about a factor of four differences in the relaxation angular dependence term. The square pores only have minor differences in the relaxation angular dependence term. While a shift of 90° of the orientation of the magnetic field to the pores does not affect the constant, the ordering of the pores will affect the constant, as will a change of orientation of the magnetic field by an angle other than 90° .

As stated, the rhombic pore system is a pedagogical example. Real systems will be further complicated because they contain a distribution of aspect ratios and complex pore shapes. Extensive experimental work coupled with simulations based on SEM and FIB-SEM images will likely be necessary to fully understand the effect and extent of homonuclear dipolar coupling on the relaxation response of organic rich shales. Simulation work will allow for sensitivity analysis to attempt to disentangle the effect of homonuclear dipolar coupling from

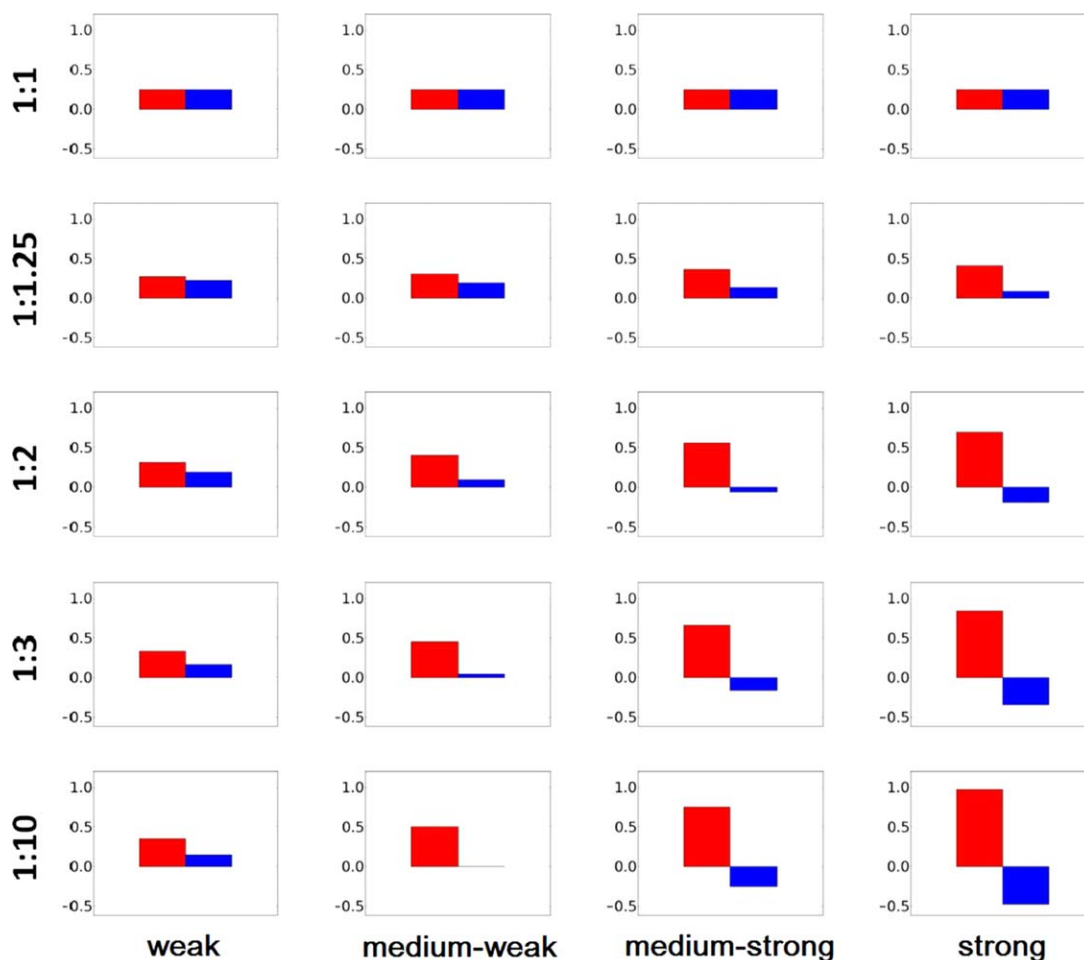


Figure 4 Relative strengths of the relaxation angular dependence term for different aspect ratios and strengths of pore ordering in the fast diffusion case. Results in red are for the long axis of the pore parallel to the applied magnetic field, results in blue perpendicular.

other relaxation mechanisms. However, in some systems, separating out the different effects may simply not be possible.

The assumption of the dipolar coupling constant being equal for all pore surfaces is also a simplification that is unlikely to reflect reality. The organic pore surfaces contain not only hydrogen but also carbon, oxygen, and nitrogen. While the heteronuclear dipolar coupling strength between hydrogen and carbon and oxygen nuclei is not greatly weaker than hydrogen homonuclear dipolar coupling, the low abundance of the NMR active isotopes makes it unlikely heteronuclear dipolar coupling is a significant factor in relaxation. Nitrogen 14 has a high isotopic abundance and a large quadrupolar moment that can lead to additional relaxation, but even for formations known to have high concentrations of nitrogen in the organic matter, such as the Green River formation, the percentage of nitrogen is still

only on the order of a few percent so that it too is unlikely to have a large contribution to relaxation. As such, it is likely that the surface relaxivity of a system would decrease with increased maturity, as the hydrogen index of kerogen decreases as it undergoes catagenesis to produce oil and gas. Provided diffusion is rapid enough, the different dipolar coupling constants in the pore space should average to a single value, just as with the paramagnetic situation. However, if some molecules have extended periods adsorbed to the surface compared to their dipolar relaxation rates, multiexponential decays may arise.

Unlike conventional reservoirs, the shale pore matrix may significantly contribute to measurable relaxation results in addition to the fluid signal. Fleury et al. (74) have looked at the hydrogen present in clays. Other work has looked at shales rich in organic solids both at low field (75) and high field (76). As expected, the relaxation times for organic solids had very short T_2

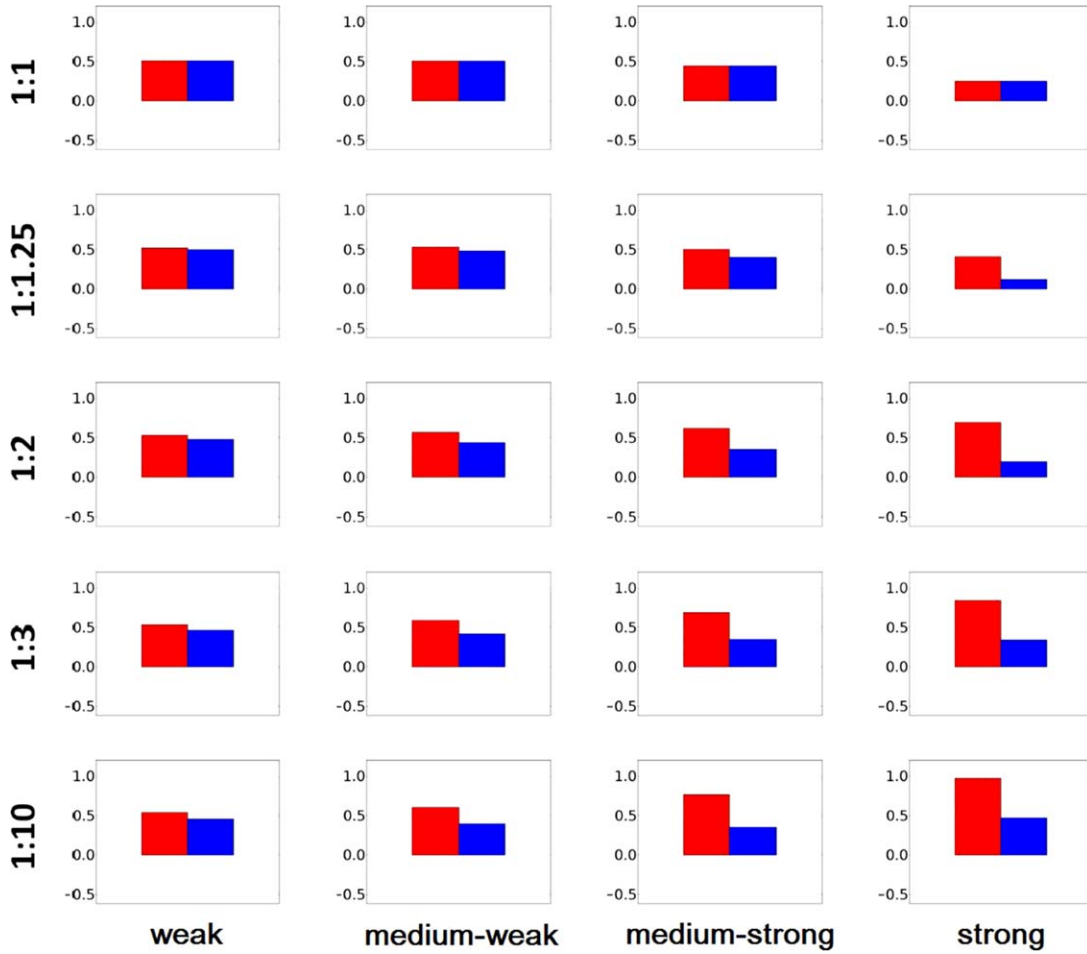


Figure 5 Relative strengths of the relaxation angular dependence term for different aspect ratios and strengths of pore ordering in the slow diffusion case. Results in red are for the long axis of the pore parallel to the applied magnetic field, results in blue perpendicular.

values and significantly longer T_1 values. However, some of the resulting T_2 relaxation times appeared to be anomalously short. The echo spacings used for the measurements were $60 \mu\text{s}$ for low field and $22 \mu\text{s}$ for the high field, and yet significant signal was seen at T_2 times on the order of $10\text{--}30 \mu\text{s}$. In particular, the measurements made with solid echoes, which will refocus transverse dephasing due to homonuclear dipolar coupling, showed significant quantities of signal at these short T_2 times. Because T_2 is the time it takes for the signal to decay away to $1/e$ of the original intensity, some signal is left after a constituent's T_2 time. While simulations showed that the inverse Laplace transform appears capable of capturing and accurately inverting some relaxation times below the echo spacing of the CPMG measurement for high SNR measurements (75), there is another possible explanation for these very short values. Transverse relaxation in liquids decays as an exponential. Solids, conversely, have transverse

relaxation that typically decays as a Gaussian, but may decay as an exponential depending on the motional regime:

$$M(t) = \sum \exp\left(-\left(\frac{t}{T_{2s}}\right)^2\right) + \sum \exp\left(-\frac{t}{T_{2l}}\right) \quad (25)$$

where T_{2s} and T_{2l} represent the T_2 values of solids and liquids, respectively. While ideally an inversion that fits to both the Gaussian and exponential portions of the signal decay would be applied for data analysis, such methods are often not possible due to instability, nonuniqueness of the solutions and the requirement for a priori knowledge of the system. Recent work by Zhu et al. (77) presents a generalized inversion to attempt to overcome these limitations, but due to its current novelty, it has not been tested on these types of systems at this time. Therefore, out of necessity, both the Gaussian and exponential portions have been typically analyzed

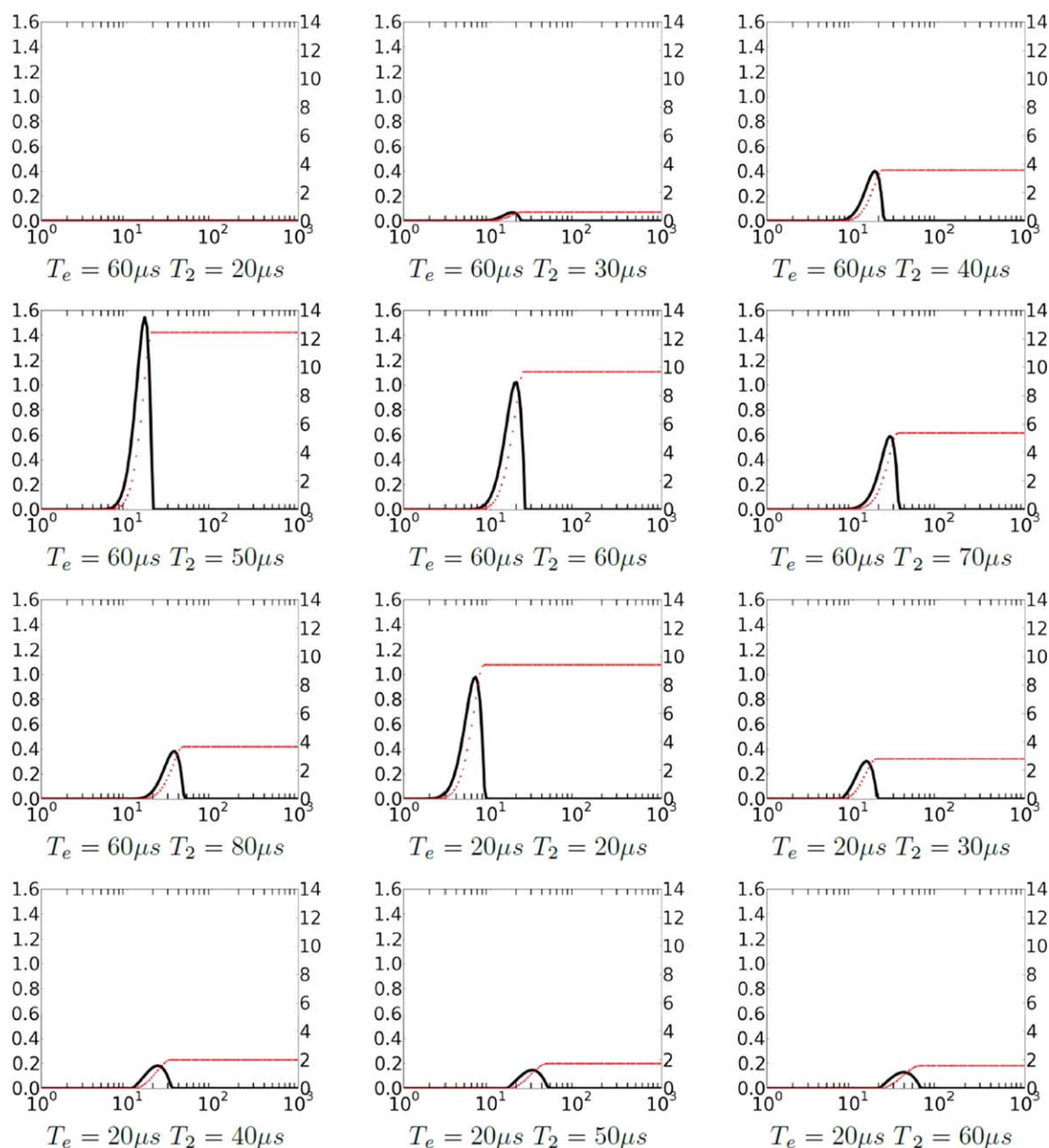


Figure 6 Gaussian decays inverted via an inverse Laplace transform for a range of echo spacings and characteristic T_2 times. Solid black lines are the incremental results and red dots are the cumulative results. All signals were simulated with an intensity of 1. Signal overcall from the inversion is present in all values of T_2 , but is most significant for relaxation values near the echo spacing.

using decaying exponentials through an inverse Laplace transform. However, when an inverse Laplace transform is applied to a Gaussian decay, the resulting relaxation time appears at erroneously low times. It should be noted, that while the inverse Laplace transform produces relaxation values for solids that are too short, the results are repeatable and can still be used to obtain useful information about the system. Figure 6

shows the effect of using an inverse Laplace transform to invert a simulated singular Gaussian decay for a range of T_2 values and using echo spacings of 60 and 20 μ s. The total signal intensity for each simulation was set to 1. A Butler-Reed-Dawson inversion (78) was used to perform the inverse Laplace transform here. This method is the common standard used in the oil industry, but other techniques for performing an

inverse Laplace transform exist and exact results will depend on the type of inversion applied to the data.

For a 60 μs echo spacing, a T_2 relaxation time of 20 μs is unrecoverable. As we move to longer T_2 times, the inversion can begin to recover the signal. Initially, there is no significant shift in the peak location, however, the intensity of the peak is increased. Because the inverse Laplace transform cannot properly fit a Gaussian decay with the sum of exponential decays, it has a tendency to over calculate the signal intensity. This effect is particularly amplified for relaxation rates near the echo spacing of the measurement. Comparing the results simulated at $T_E = 20 \mu\text{s}$ and $T_E = 60 \mu\text{s}$, increasing the sampling rate appears to mitigate the overcall. This may in part explain some of the signal discrepancies seen by Washburn et al. (76) between their low and high-field results, where the low-field measurements produced stronger peaks at short relaxation times relative to the high-field results. These results indicate we need to be concerned not only about different possible relaxation mechanisms in shales but also potential artifacts from processing the data using the standard inverse Laplace transform methods used for conventional petroleum reservoirs.

RESIDUAL DIPOLAR COUPLING

In addition to homonuclear dipolar coupling between liquids and organic pore surfaces, dipolar coupling may have other effects on the fluid. It has been found that the rotational diffusion coefficient of fluid molecules can be rendered anisotropic if there are larger particles in solution that are aligned. The alignment will lead to some rotations that are more favorable than others. Unlike an isotropically diffusing system where the dipolar coupling will average out to zero, because some values of θ will be more likely than others in an anisotropically constrained system, there will be incomplete orientational averaging resulting in a residual dipolar coupling, which will then lead to additional dephasing of the transverse relaxation.

For conventional samples, the pore space is assumed to be roughly spherical. This obviously is a simplification, however, the pore space is considered to be mostly isotropic. In contrast, the pores in many shales are an often highly anisotropic slit shaped. These types of systems can produce anisotropy in the diffusion coefficient; the molecules are more free to diffuse along the major axis of the pore space than the minor axis. For small molecules, this alone is unlikely to have a significant effect on the rotational diffusion coefficient. However, for longer chain hydrocarbons, this may impede rotation in the direction of the minor axis of the pore. Valiullin and

Khokhlov (79) presented work on linear alkanes in mesoporous silica nanotubes. In this situation, the anisotropic diffusion in the pore space led to residual dipolar couplings on the order of 200–300 Hz. The pores of that study were on the order of 6 nm. While smaller than many pores in shales, recent FIB-SEM work suggests that there exist systems where the organic porosity is on the order of that size. It is possible that samples may exist where the residual dipolar coupling is the major source relaxation, but in general, it would not be expected that it would be a significant factor in the relaxation time for most samples based on the results of Valiullin.

However, an important difference exists between the silica nanotubes and the porosity found in shales. It is the hindering of the rotational diffusion coefficient, not the translational diffusion coefficient that leads to the development of residual dipolar couplings. In the case of silica nanotubes, due to wettability, there is not expected to be significant interaction between the n-alkanes in the fluid and the surface, such that pores needed to be very small to have an effect on the rotational diffusion coefficient of the molecules. A better analogy may be the anisotropic diffusion that occurs in biological materials. Many biological systems, such as tendons and muscles, contain ordered structures. The adsorption of water molecules on the surfaces and exchange between the free and bound populations leads to hindered rotation and produces significant residual dipolar couplings despite having diameters on the orders of a few microns. In these systems, residual dipolar coupling constants of up to a kilohertz have been observed (80–83). Because the residual dipolar coupling is also dependent on the orientation of the applied magnetic field, this has led to observation of the transverse relaxation time containing an orientation dependence in samples such as tendons and muscle (72). In the fluid filled organic pores, there may be significant adsorption of the gases and fluids on the pore surface depending on the system, impeding isotropic rotational motion of the molecules. Fluids may also adsorb to the surface of clay particles (84,85). These could lead to further incomplete orientational averaging of the dipolar coupling. Therefore, residual dipolar couplings have the potential to arise in larger pores in shales. This effect is most likely to be prominent in shorter chain molecules, as longer chain molecules are better able to reorient themselves. As with the homonuclear dipolar coupling between liquids and solids in the shales, the residual dipolar coupling again raises the possibility of orientational dependence of the NMR relaxation in shales. Nonetheless, due to the relatively weak strength of residual dipolar couplings, they are likely to be a secondary source of relaxation in shales as opposed to a main factor.

MAGNETIZATION TRANSFER

Another effect that may influence relaxation results in shales is magnetization transfer. Here, there are two or more populations of spins undergoing relaxation at different rates. Due to interactions between these environments, there is a swapping of magnetization that leads to an averaging of the relaxation times. In shales, magnetization exchange can potentially arise through a variety of mechanisms.

When two bonded spins experience dipolar coupling, an exchange of magnetization between two spins may occur, called spin diffusion (86,87). The term “spin diffusion” is often used when molecular diffusion is what is meant. Molecular diffusion is the physical movement of a molecule, whereas spin diffusion is the exchange of magnetization between two coupled nuclei. Because the nuclei are exchanging magnetization, this can lead to averaging of relaxation times between domains where the nuclei are relaxing at different rates. This is common in highly coupled systems, such as proteins, leading to a single relaxation value for the whole molecule. Spin diffusion can be both intramolecular and intermolecular, though some researchers reserve the term for intramolecular interactions and use the more general term of magnetization transfer for intermolecular interactions.

Spin diffusion is slow compared to molecular diffusion, with coefficients on the order of a $1\text{--}10\text{ nm}^2/\text{ms}$. However, given the long T_1 of organic solids even at low magnetic fields (75) and the small pore sizes found in shales (47), the distance traveled by the diffusion of magnetization can easily become on the order of some of the pore sizes before relaxing. In addition, the small pore sizes have extremely high surface to volume ratios, giving a large exposed surface for the liquids and solids to interact.

Another interaction that may cause magnetization transfer is chemical exchange (88–91). Here, molecular dissociation will lead to a physical swap of the hydrogen atoms between two environments. The likelihood of chemical exchange depends on the bond dissociation energies in both the organic matter and the fluids present. As organic matter loses acid groups during the production of oil and gas, it is expected that chemical exchange will be more likely in samples of lower thermal maturity. However, due to high dissociation energies of hydrocarbons, chemical exchange is not expected to be a significant factor in magnetization transfer in organic rich shales. The nuclear Overhauser effect may also lead to polarization transfer in shales through cross-relaxation (91–93). Because the nuclear Overhauser effect is through space, it is expected to be a weaker

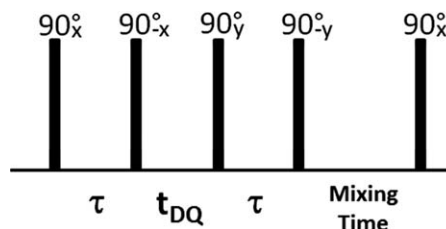


Figure 7 Double quantum filter spin-diffusion pulse sequence.

interaction than for dipolar interactions between chemically bonded spins.

To test whether magnetization transfer was occurring between solids and liquids in shale samples, we performed a basic double-quantum filter measurement and double-quantum filtered spin diffusion measurements to monitor the movement of magnetization. A double quantum filter is used to filter out the signal arising from mobile phases like liquids, such that only the signal from immobile phases remains (94). The magnetization then can be stored for a mixing time, allowing exchange of magnetization between different populations of spins. A measurement is then taken to observe how the magnetization has moved during the mixing period. The double quantum filtered spin diffusion pulse sequence is shown in Fig. 7 (95). The first pulse excites the system, which is allowed to evolve in the presence of dipolar coupling for a time τ . The second pulse converts the magnetization into double quantum coherences, which evolve during the time t_{DQ} . As double quantum coherences cannot be observed directly, a third pulse is applied to convert them back into a single quantum coherence. A pair of 90° pulses are used to store the magnetization for a mixing time to allow movement of magnetization between different domains. Pulse length was $12\text{ }\mu\text{s}$ and both τ and t_{DQ} were $30\text{ }\mu\text{s}$. Mixing times varied from 30 to $1,000\text{ }\mu\text{s}$. Measurements were made on a native state, organic rich tight oil shale sample and an immature oil shale from the Glen Davis mine in Australia. The tight oil sample contains a significant amount of oil and water, whereas the immature oil shale sample contains only organic solids. Measurements were made on a Magritek 2 MHz Rock Core Analyser using a P35 probe optimized for shales.

When a basic double quantum filter is performed on the shale samples, we see a broad peak consistent with what one would expect from solids for both samples, shown in Figs. 8(a) and 9(a). If we allow a $30\text{ }\mu\text{s}$ mixing time, we see a significant, narrow peak appear in the spectrum of the tight oil sample, Fig. 8(b). The narrow peak comes from molecules not experiencing significant broadening from dipolar coupling and its appearance in the spectra must arise from movement of magnetization

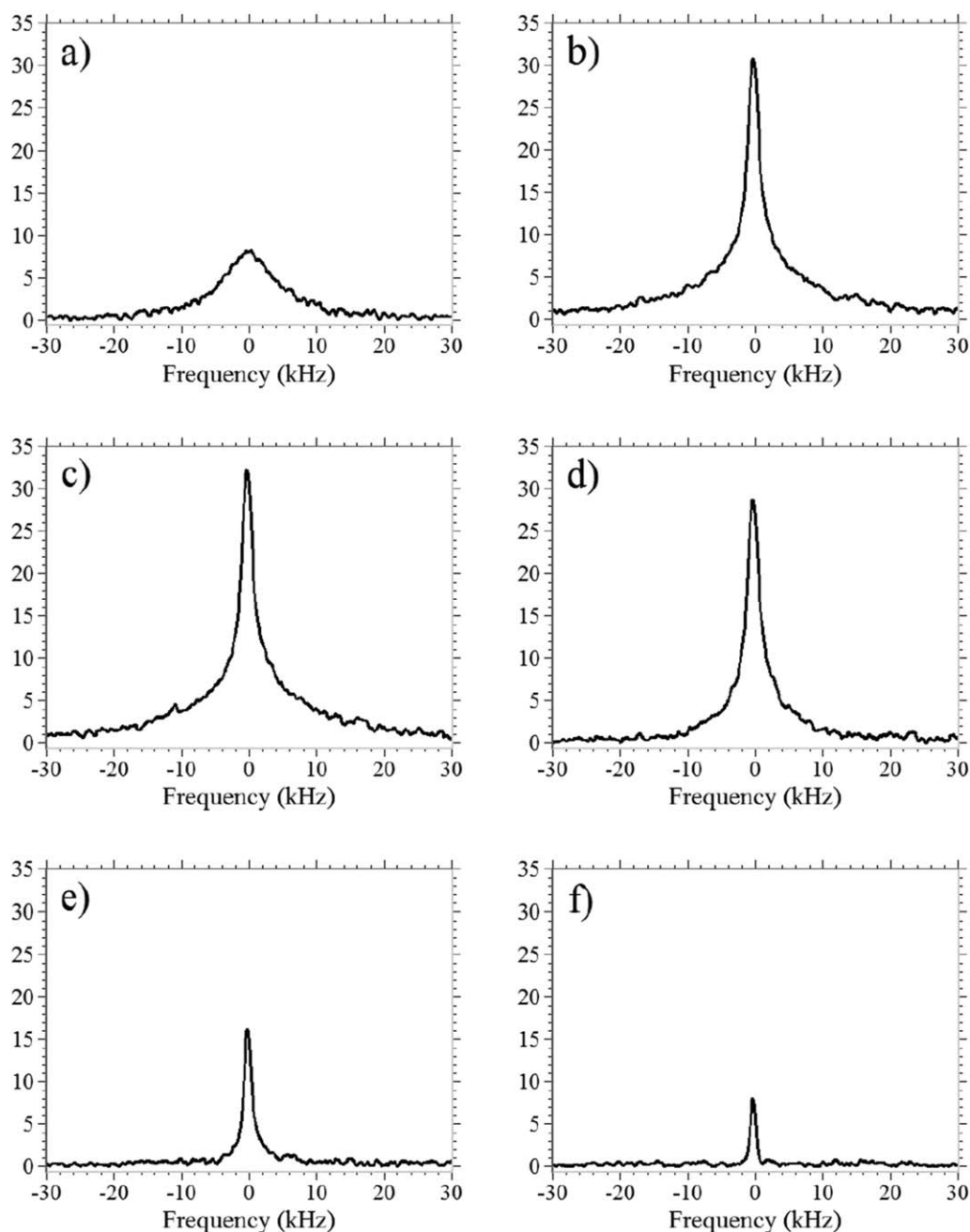


Figure 8 Measurement of magnetization transfer in a native state tight oil sample. (a) No mixing time, (b) 30 μ s mixing time, (c) 50 μ s mixing time, (d) 100 μ s mixing time, (e) 500 μ s mixing time, and (f) 1,000 μ s mixing time.

from an immobile phase to a mobile phase. This is likely due, at least in part, to movement of magnetization from the organic solids to saturating liquids, but may be due to transfer of magnetization from viscous solids to less viscous solids or clays. Unfortunately, due to the acquisition limits of the equipment, a significant portion of the solid signal does not appear to be captured due to its short T_2^* times and will not appear in the

spectra. Based on the results, there is almost certainly signal from solids beyond what we can measure in the spectra that interacts with mobile portions in the sample. Future work with equipment with shorter dead times should allow a more complete description of the system response, allowing a better understanding how magnetization moves between the shale constituents. As mixing time is increased, a small increase in

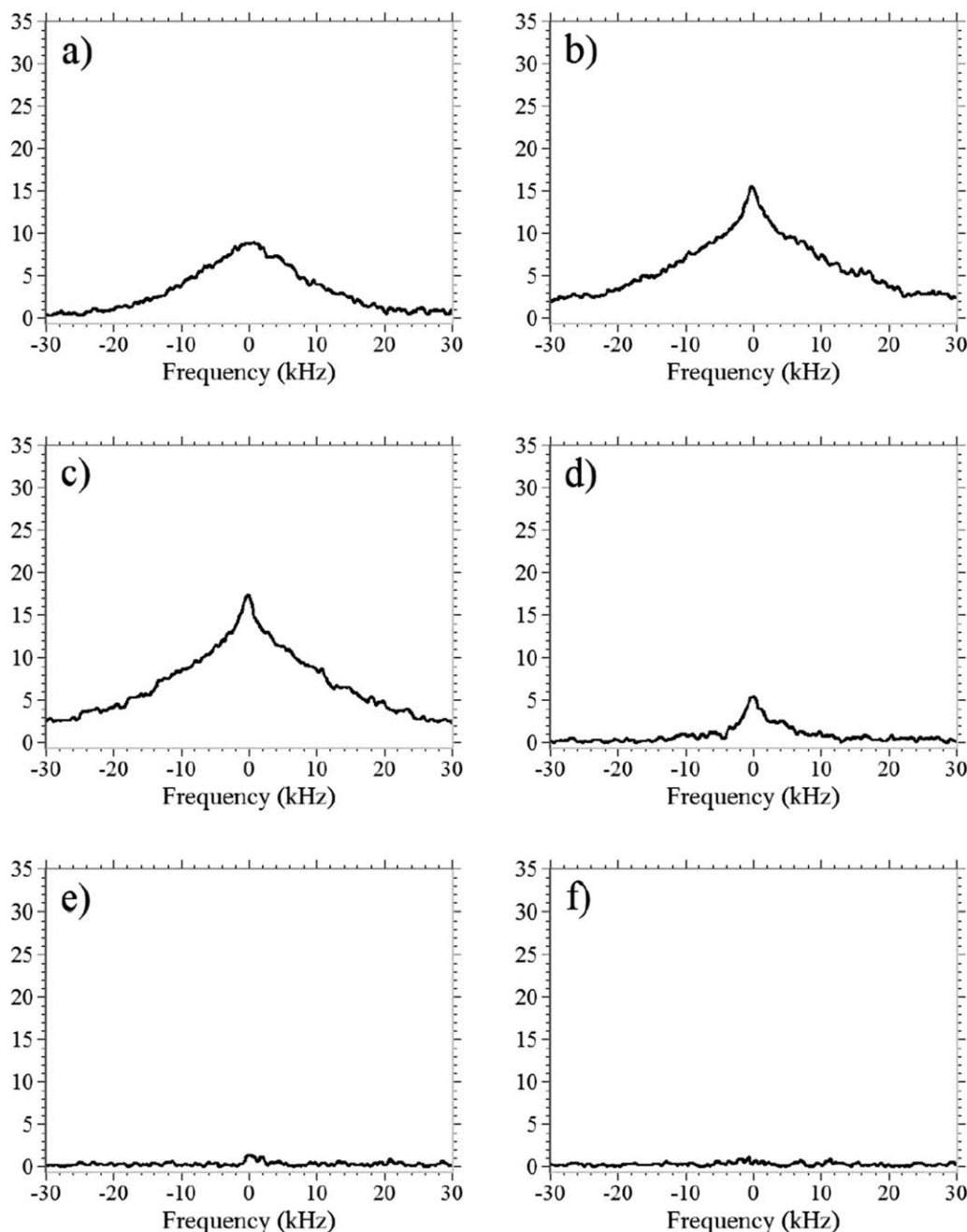


Figure 9 Measurement of magnetization transfer in an oil shale sample. (a) No mixing time, (b) 30 μ s mixing time, (c) 50 μ s mixing time, (d) 100 μ s mixing time, (e) 500 μ s mixing time, and (f) 1,000 μ s mixing time.

intensity is initially observed and then, while the total signal intensity decreases, one can observe a narrowing of the linewidth, consistent with movement of magnetization from solid to liquid components. In contrast, the oil shale sample only shows a slight increase in magnetization and a slight sharpening of the peak, Fig. 9(b), before decaying away with increased mixing time. This may stem from domains of differing viscos-

ity of kerogen in the sample, which agrees with the high-field NMR work on the oil shale (76), where two domains were seen in the T_1 dimension.

These results indicate that significant movement of magnetization between immobile to mobile phases occurs in just the span of tens of microseconds. Given the organic solids in the samples can have T_1 relaxation times on the order of hundreds or thousands of

milliseconds (75), this suggests the potential for the unusual situation where the interaction of liquids with the organic solids may actually lead to an increase in the longitudinal relaxation time of the liquid. This behavior was also observed by Valiullin and Furo (96) in nanoporous ice, where the exchange of magnetization between solid ice with a long T_1 and liquid water in nanopores with a short T_1 led to a single averaged longitudinal relaxation time. In addition, work of Valiullin and Furo in nanopores in ice indicates that the process of magnetization exchange can be enhanced by molecular diffusion; while spin diffusion is limited to nearby nuclei in solids, exchange of fluid molecules between the surface and bulk will provide fresh spins for exchange.

This also raises questions regarding what effect the presence of pyrite in framboids in the samples will have on the relaxation signal. It has been demonstrated that paramagnetic impurities need to be at the surface to have a large impact on the relaxation time of fluids (30). Therefore, while the large framboids are unlikely to act significantly as a source of surface relaxivity, they could act as paramagnetic sinks for relaxation through spin diffusion from adjacent organic solids. Exchange of magnetization between solids and saturating fluids will lead to an averaging of relaxation times. Therefore, framboids may produce more significant relaxation effects in organic rich samples than in conventional inorganic samples. For the Barnett samples examined by Sondergeld and coworkers (97) with surface relaxivities on the order of 50 $\mu\text{m/s}$, SEM images of Barnett samples showed that much of the kerogen in the sample was adjacent to pyrite framboids.

CONCLUSIONS

Interactions that can be neglected in conventional porous systems, such as spin diffusion, chemical exchange, the nuclear Overhauser effect, homonuclear dipolar coupling and residual dipolar coupling may have a significant influence on the measured relaxation signal in unconventional resources such as shales. Analysis of relaxation results using the traditional inverse Laplace transform may lead to anomalously low relaxation values and overcalled intensity for Gaussian signals arising from solids. Diffusional coupling may lead to relaxation distributions that reflect not the pore size distribution, but an average of that pore size distribution with dependence on the pore shapes and tortuosity of the system. Interpretation of the NMR response may be further hindered by different relaxation rates between the organic and inorganic porosity. In contrast, effects such as internal gradients

that confound relaxation measurements in conventional samples are unlikely to have a significant impact in shales. The small pore sizes also indicate that applied gradients from logging tools and benchtop instruments are unlikely to be able to measure the diffusion coefficient of fluids in many shales.

This article is not intended to be a definitive guide but a starting point for further experimental and simulation work. Given the wide range of shale compositions, structures, and maturities, it is expected that some of the mechanisms presented may play a significant role for some systems and be negligible for others. The importance of the differing mechanisms are dependent on the surface residence time of adsorbed fluids and gases, which at present time are still poorly understood. As more information is available, both from NMR and other types of measurements, our understanding of the relative importance of the different mechanisms will improve.

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